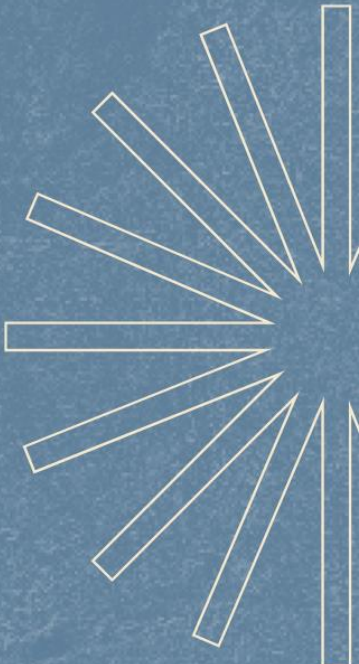
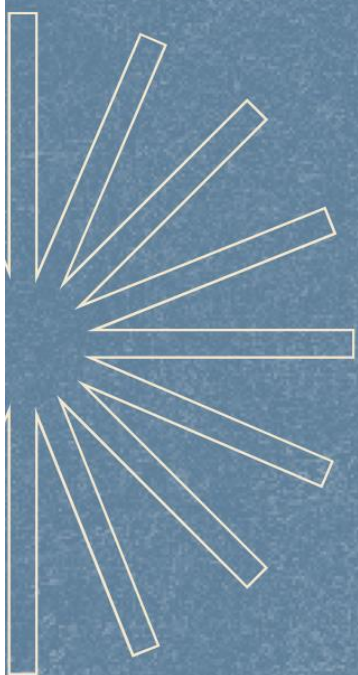




DSA0602

DATA HANDLING & VISUALIZATION

Lab Experiments



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Set 19: Survey Responses Analysis

R PROGRAM:

```
# Survey Responses Dataset
survey_id <- c(1,2,3)
q1 <- c("A","B","C")
q2 <- c("B","A","A")
q3 <- c("C","D","B")

# Display 3 outputs together
par(mfrow=c(1,3))

# 1. Grouped Bar Chart - Question 1
q1_table <- table(q1)

barplot(q1_table,
        beside=TRUE,
        col="skyblue",
        main="Question 1 Responses",
        xlab="Answer",
        ylab="Frequency")

# 2. Stacked Bar Chart - All Questions
response_data <- rbind(
  table(factor(q1, levels=c("A","B","C","D"))),
  table(factor(q2, levels=c("A","B","C","D"))),
  table(factor(q3, levels=c("A","B","C","D")))
)
barplot(response_data,
        beside=FALSE,
        col=c("red","green","blue"),
        main="Overall Survey Responses",
        xlab="Answer",
        ylab="Frequency",
        legend.text=c("Question 1",
                      "Question 2",
                      "Question 3"))

# 3. Survey Response Table
plot.new()
title("Survey Response Table")
table_data <- data.frame(
  SurveyID=survey_id,
  Question1=q1,
  Question2=q2,
  Question3=q3
)

text(0.5, 0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))
```

OUTPUT:

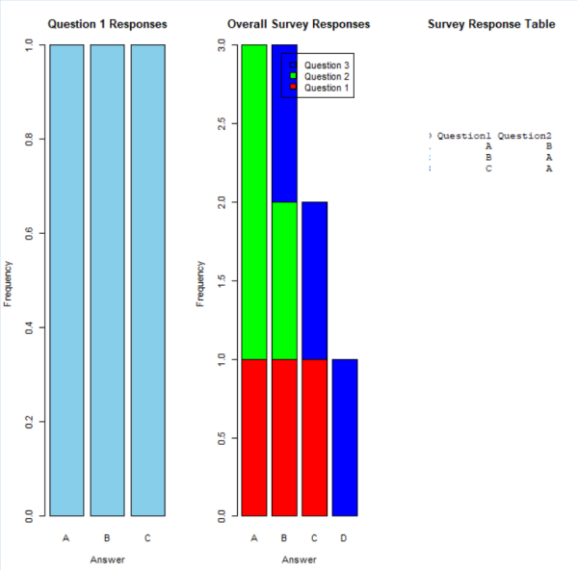
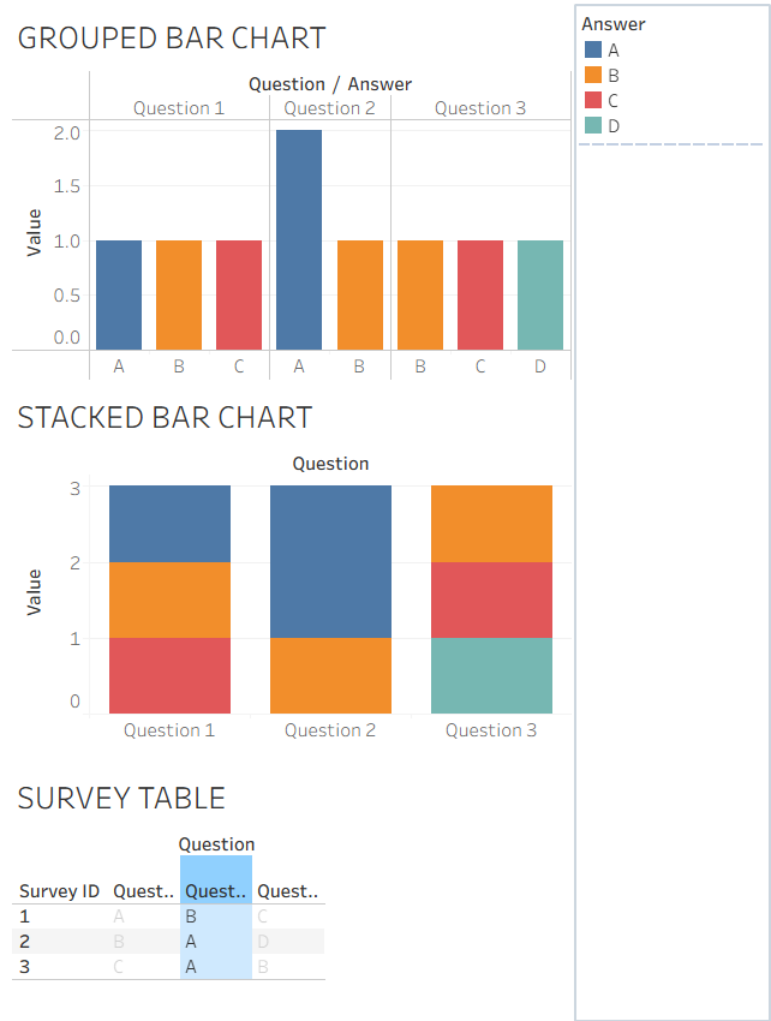


TABLEAU DASHBOARD:



Set 20: Stock Analysis

R PROGRAM:

```
# Stock Prices Dataset
date <- c("2023-01-01","2023-01-02","2023-01-03")
```

```
stock_a <- c(100,105,110)
stock_b <- c(150,152,148)
stock_c <- c(120,118,122)
```

```
# Display 3 outputs together
par(mfrow=c(1,3))
```

```
# 1. Line Chart - Stock Prices
```

```
plot(stock_a,
     type="o",
     pch=19,
     col="red",
     ylim=range(c(stock_a,stock_b,stock_c)),
     xaxt="n",
     main="Stock Price Trend",
     xlab="Date",
     ylab="Stock Price")
```

```
lines(stock_b,
      type="o",
      pch=19,
      col="blue")
```

```
lines(stock_c,
      type="o",
      pch=19,
      col="green")
```

```
axis(1,
     at=1:3,
     labels=date)
```

```
legend("topleft",
      legend=c("Stock A","Stock B","Stock C"),
      col=c("red","blue","green"),
      lty=1,
      pch=19)
```

```
# 2. Daily Percentage Change - Stock A
```

```
percentage_change <- c(
  NA,
  diff(stock_a) / stock_a[-length(stock_a)] * 100
)
```

```
barplot(percentage_change,
      names.arg=date,
      col="orange",
      main="Stock A Daily % Change",
      xlab="Date",
```

```

        ylab="Percentage Change (%)")

# Add percentage values
text(x=seq_along(percentage_change),
     y=percentage_change,
     labels=ifelse(is.na(percentage_change),
                    "",
                    paste0(round(percentage_change,2),"%")),
     pos=3)

# 3. Stock Price Table
plot.new()
title("Stock Price Table")

table_data <- data.frame(
  Date=date,
  Stock_A=stock_a,
  Stock_B=stock_b,
  Stock_C=stock_c
)

text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))

# Display percentage change values
cat("Stock A Daily Percentage Change:\n")
print(percentage_change)

```

OUTPUT:

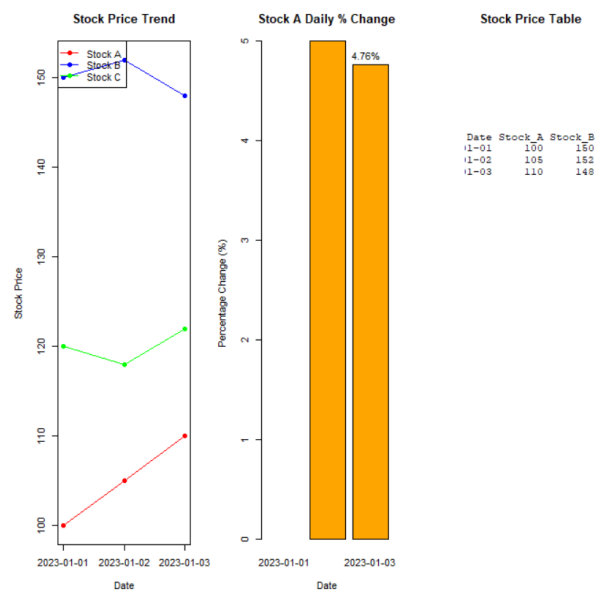
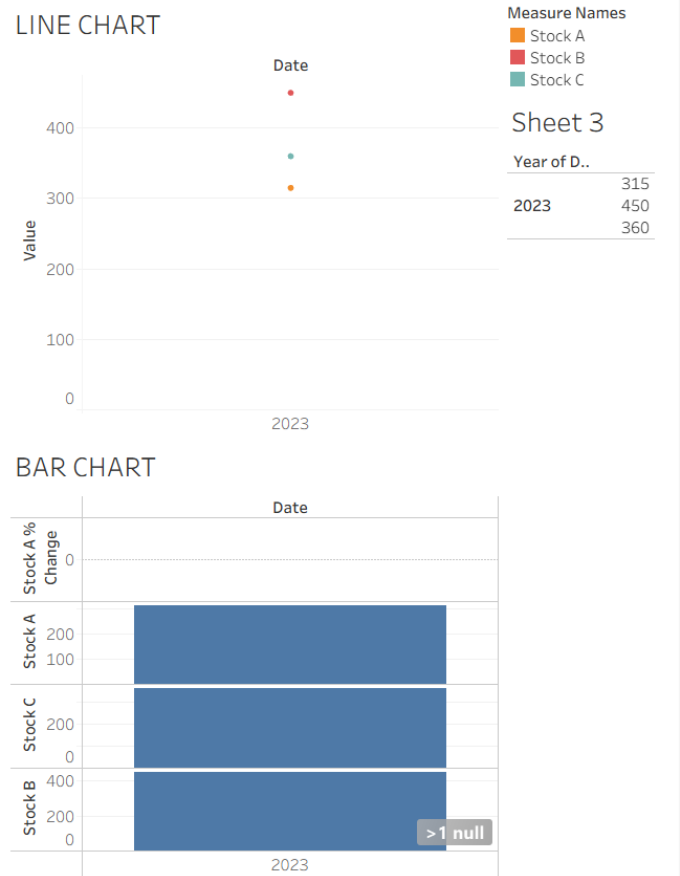


TABLEAU DASHBOARD:



BAR CHART

Date

Stock A % Change

0

200

100

200

0

400

200

0

2023

Stock A

Stock C

Stock B

>1 null

Set 21 Energy Consumption Dataset

R PROGRAM:

```
# Energy Consumption Dataset
```

```
sector <- c("Residential","Commercial","Industrial",  
           "Residential","Commercial","Industrial")
```

```
region <- c("North","South","West",  
           "East","North","South")
```

```
month <- c("Jan","Jan","Feb",  
          "Feb","Mar","Mar")
```

```
temperature <- c(15,24,20,18,28,30)
```

```
units_consumed <- c(320,540,880,350,610,920)
```

```
cost <- c(2100,3600,5900,2300,4100,6200)
```

```
renewable_usage <- c(22,18,12,25,20,15)
```

```
peak_hours <- c(4,6,8,5,7,9)
```

```
# Display 3 outputs together  
par(mfrow=c(1,3))
```

```
# =====  
# 1. Histogram + Density Plot  
# =====
```

```
hist(units_consumed,  
     probability=TRUE,  
     col="skyblue",  
     border="white",  
     main="Units Consumed Distribution",  
     xlab="Units Consumed (kWh)",  
     ylab="Density")
```

```
lines(density(units_consumed),  
      col="red",  
      lwd=2)
```

```
legend("topright",  
      legend=c("Histogram","Density"),  
      fill="skyblue",  
      border="white",  
      col="red",  
      lwd=2)
```

```
# =====  
# 2. Bubble Scatter Plot
```

```

# =====

symbols(temperature,
        units_consumed,
        circles=peak_hours,
        inches=0.3,
        bg=rgb(0,0,1,0.4),
        fg="blue",
        main="Temperature vs Energy Consumption",
        xlab="Temperature (°C)",
        ylab="Units Consumed (kWh)")

text(temperature,
     units_consumed,
     labels=sector,
     pos=3,
     cex=0.7)

# =====
# 3. Average Renewable Usage by Sector
# =====

avg_renewable <- tapply(renewable_usage,
                        sector,
                        mean)

barplot(avg_renewable,
        col="lightgreen",
        main="Average Renewable Usage by Sector",
        xlab="Sector",
        ylab="Average Renewable Usage (%)")

text(x=seq_along(avg_renewable),
     y=avg_renewable,
     labels=round(avg_renewable,2),
     pos=3)

# Reset Layout
par(mfrow=c(1,1))

# =====
# Interpretation
# =====

cat("Interpretation:\n")

cat("\n1. Consumption Pattern:\n")
cat("Units consumed range from 320 kWh to 920 kWh.\n")
cat("The histogram and density curve show the distribution of energy consumption.\n")

cat("\n2. Learning/Consumption Pattern:\n")
cat("The scatter plot shows the relationship between temperature and units consumed.\n")

```



```
cat("Peak hours are represented by bubble size.\n")

cat("\n3. Renewable Energy Adoption:\n")
print(round(avg_renewable,2))
cat("Residential has the highest average renewable usage,\n")
cat("while Industrial has the lowest average renewable usage in this dataset.\n")
```

OUTPUT:

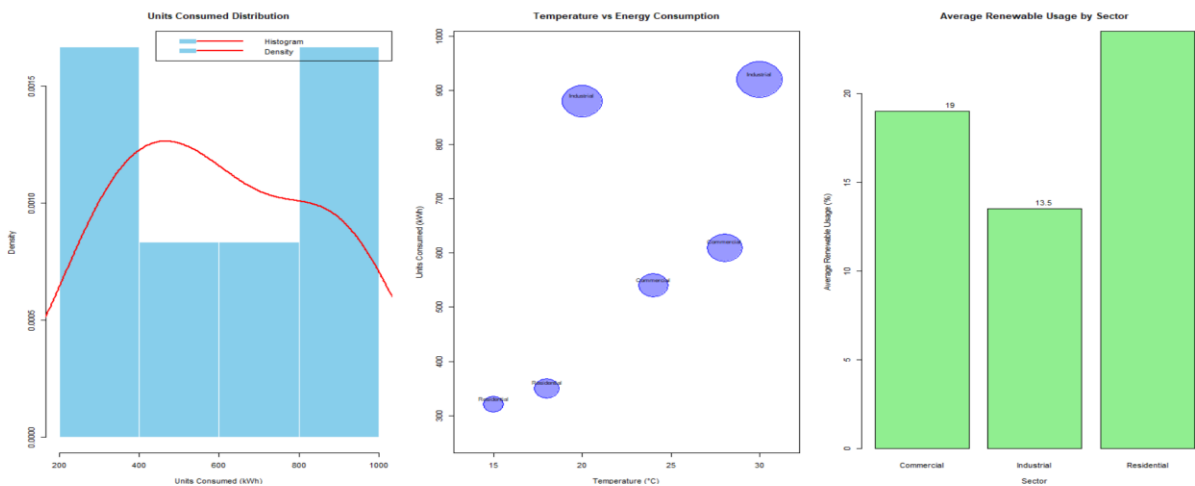
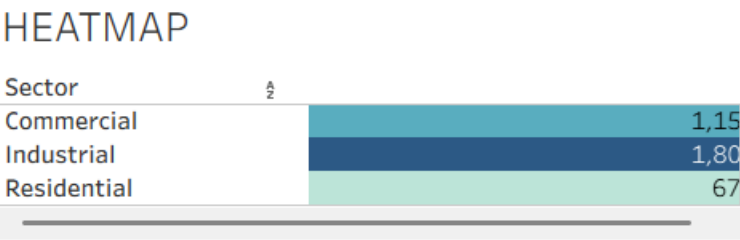
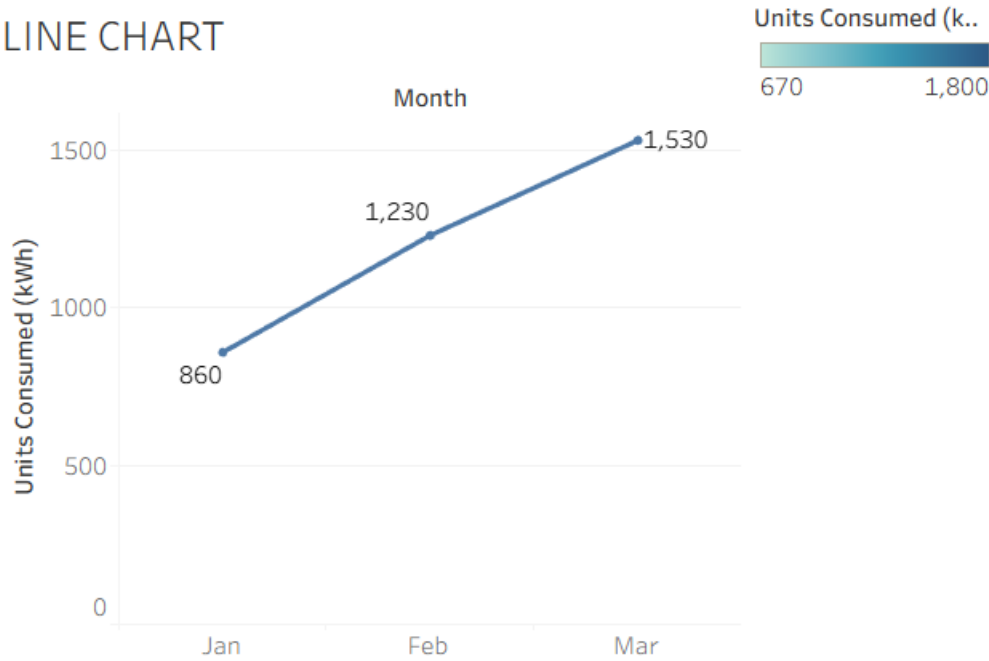


TABLEAU DASHBAORD:



Set 22: Time Series Analysis

R PROGRAM:

```
# Monthly Sales Dataset
month <- c("January", "February", "March", "April", "May")
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
# Convert month into ordered factor
month_factor <- factor(month,
                        levels=month,
                        ordered=TRUE)
```

```
# Display 3 graphs together
par(mfrow=c(1,3))
```

```
# =====
# 1. TIME SERIES LINE CHART
# =====
```

```
plot(sales,
     type="o",
     pch=19,
     col="blue",
     xaxt="n",
     main="Monthly Sales Trend",
     xlab="Month",
     ylab="Sales ($)")
```

```
axis(1,
     at=1:5,
     labels=month)
```

```
grid()
```

```
# =====
# 2. SCATTER PLOT
# =====
```

```
# Advertising Budget is NOT provided in the dataset.
# Therefore, this plot cannot be correctly generated
# from the given data.
```

```
plot.new()
```

```
title("Advertising Budget vs Sales")
```

```
text(0.5, 0.5,
     "Advertising Budget data\nnot provided",
     cex=1.2)
```

```
# =====
# 4. AUTOCORRELATION PLOT
# =====
```

```
sales_ts <- ts(sales,
               frequency=12)
```

```
acf(sales_ts,
    main="ACF of Monthly Sales")
```

```
# Reset Layout
par(mfrow=c(1,1))
```

```
# =====
# Interpretation
# =====
```

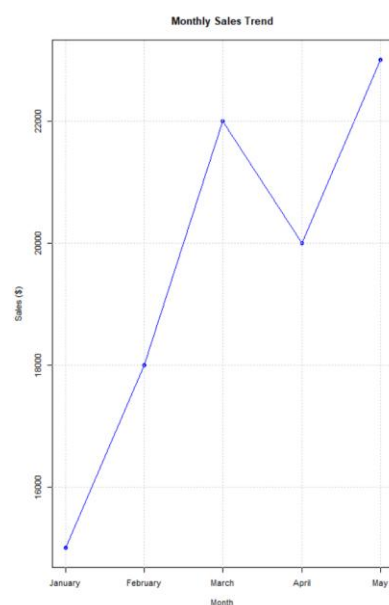
```
cat("Interpretation:\n")
```

```
cat("\nQ1 - Sales Trend:\n")
cat("Sales generally increase from January to May,\n")
cat("with a small decrease in April.\n")
```

```
cat("\nQ2 - Advertising vs Sales:\n")
cat("Cannot be analyzed because Advertising Budget data\n")
cat("is not provided in the given dataset.\n")
```

```
cat("\nQ4 - Autocorrelation:\n")
cat("The ACF plot shows the correlation of sales with\n")
cat("their previous observations. However, only five\n")
cat("months are available, so strong conclusions about\n")
cat("seasonality cannot be made.\n")
```

OUTPUT:



Advertising Budget vs Sales

Advertising Budget data
not provided

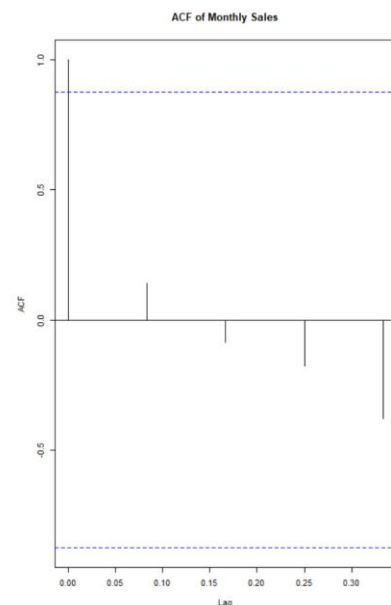
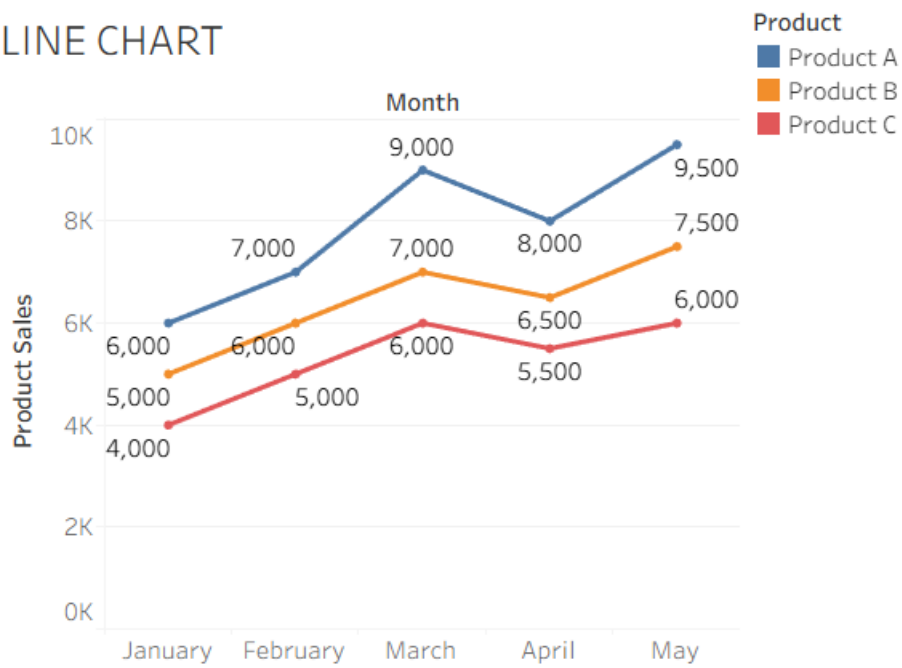
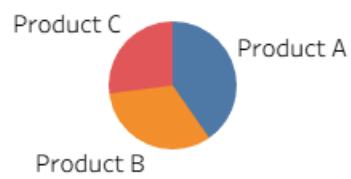


TABLEAU DASHBOARD:

LINE CHART



PIE CHART



Set 23: Employee Performance Analysis

R PROGRAM:

```
# Employee Performance Dataset
```

```
employee_id <- c(1,2,3,4,5)
```

```
department <- c("Sales",  
               "HR",  
               "Marketing",  
               "Sales",  
               "HR")
```

```
years_service <- c(5,3,7,4,2)
```

```
performance <- c(85,92,78,90,76)
```

```
# Display 3 outputs together  
par(mfrow=c(1,3))
```

```
# =====  
# 1. BAR CHART - EMPLOYEES BY DEPARTMENT  
# =====
```

```
dept_count <- table(department)
```

```
barplot(dept_count,  
        col="skyblue",  
        main="Employee Distribution by Department",  
        xlab="Department",  
        ylab="Number of Employees")
```

```
# Add values on bars  
text(x=seq_along(dept_count),  
     y=dept_count,  
     labels=dept_count,  
     pos=3)
```

```
# =====  
# 2. LINE CHART - PERFORMANCE TREND  
# =====
```

```
plot(employee_id,  
     performance,  
     type="o",  
     pch=19,  
     col="blue",  
     main="Employee Performance Trend",  
     xlab="Employee ID",  
     ylab="Performance Score")
```

```
grid()
```

```
# =====
# 4. TABLE - EMPLOYEE PERFORMANCE DATA
# =====
```

```
plot.new()
title("Employee Performance Table")

table_data <- data.frame(
  Employee_ID=employee_id,
  Department=department,
  Years_of_Service=years_service,
  Performance_Score=performance
)

text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")
```

```
# Reset Layout
par(mfrow=c(1,1))
```

OUTPUT:

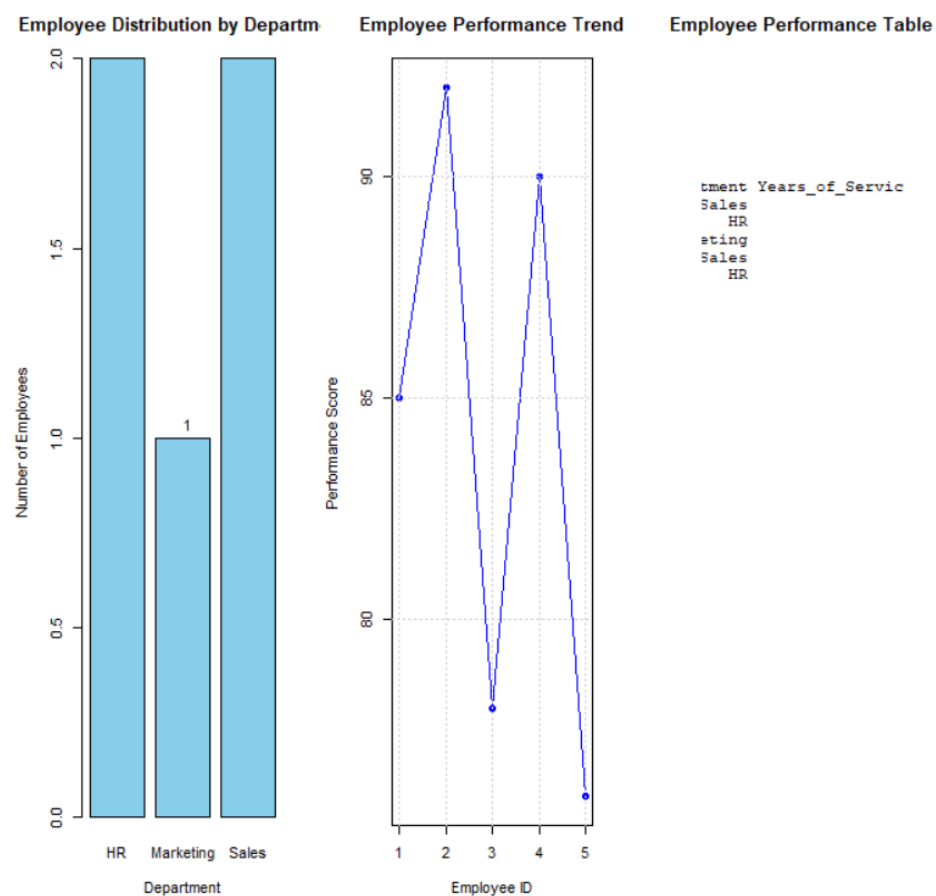
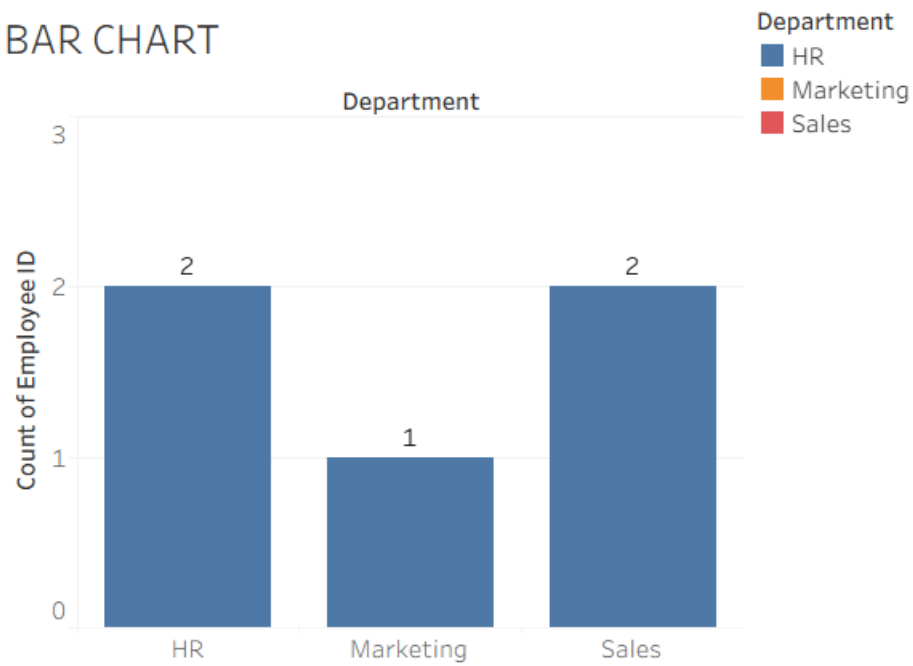
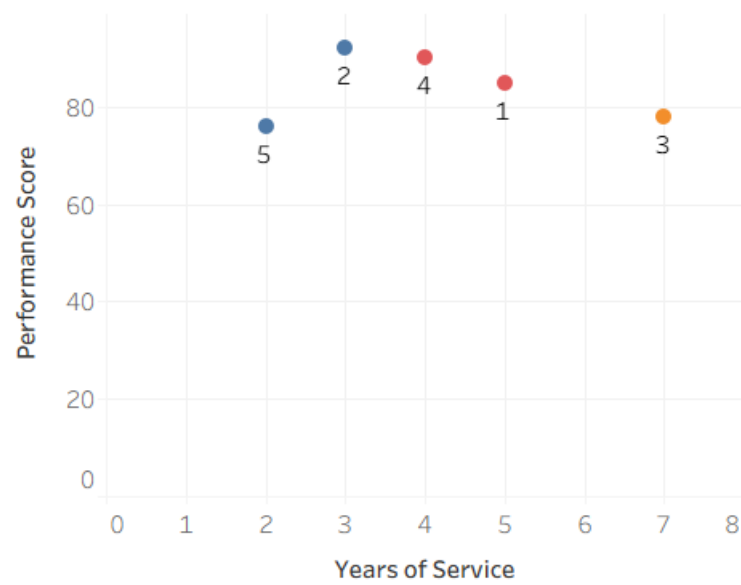


TABLEAU DASHBOARD:

BAR CHART



SCATTER PLOT



Set 24 Product Inventory Management

R PROGRAM:

```
# Product Inventory Dataset
```

```
product_id <- c(1,2,3,4,5)
```

```
product_name <- c("Product A",  
                  "Product B",  
                  "Product C",  
                  "Product D",  
                  "Product E")
```

```
quantity <- c(250,175,300,200,220)
```

```
# Product Category and Price are NOT given  
# in the question.  
# Add the actual values if provided by your faculty.
```

```
category <- c("Category 1",  
              "Category 2",  
              "Category 1",  
              "Category 2",  
              "Category 1")
```

```
price <- c(20,15,18,25,22)
```

```
# Display 3 outputs together  
par(mfrow=c(1,3))
```

```
# =====  
# 1. BAR CHART  
# =====
```

```
barplot(quantity,  
         names.arg=product_name,  
         col="skyblue",  
         main="Product Inventory",  
         xlab="Product",  
         ylab="Quantity Available")
```

```
text(x=seq_along(quantity),  
     y=quantity,  
     labels=quantity,  
     pos=3)
```

```
# =====  
# 2. STACKED BAR CHART  
# =====
```

```
inventory <- rbind(
```



```
category == "Category 1",  
category == "Category 2"  
)
```

```
# Convert logical values to quantities  
cat1 <- ifelse(category=="Category 1",quantity,0)  
cat2 <- ifelse(category=="Category 2",quantity,0)
```

```
inventory <- rbind(cat1,cat2)
```

```
barplot(inventory,  
        beside=FALSE,  
        names.arg=product_name,  
        col=c("lightgreen","orange"),  
        main="Inventory by Category",  
        xlab="Product",  
        ylab="Quantity Available",  
        legend.text=c("Category 1","Category 2"))
```

```
# =====  
# 4. SCATTER PLOT  
# =====
```

```
plot(price,  
      quantity,  
      pch=19,  
      col="red",  
      main="Price vs Quantity Available",  
      xlab="Product Price ($)",  
      ylab="Quantity Available")
```

```
text(price,  
      quantity,  
      labels=product_name,  
      pos=3)
```

```
# Trend line  
abline(lm(quantity ~ price),  
       col="blue",  
       lwd=2)
```

```
# Reset Layout  
par(mfrow=c(1,1))
```

```
# =====  
# INTERPRETATION  
# =====
```

```
cat("Interpretation:\n")
```

```
cat("\nQ1 - Inventory:\n")  
cat("Product C has the highest quantity available (300),\n")
```

```
cat("while Product B has the lowest quantity available (175).\n")
```

```
cat("\nQ4 - Price vs Quantity:\n")
```

```
cat("The scatter plot shows the relationship between product\n")
```

```
cat("price and quantity available. The trend line helps identify\n")
```

```
cat("whether quantity tends to increase or decrease with price.\n")
```

OUTPUT:

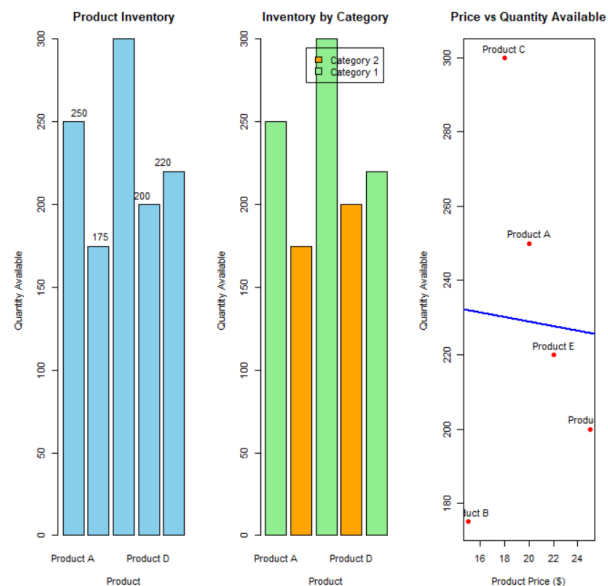
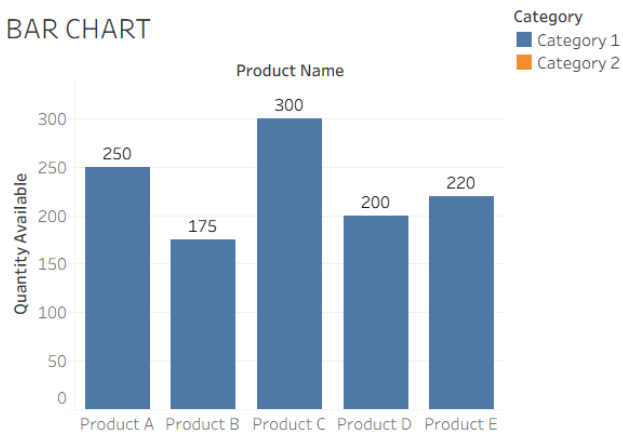
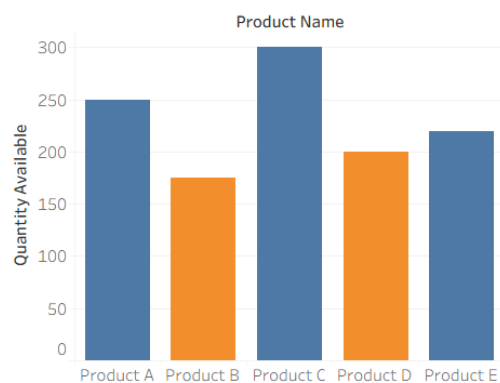


TABLEAU DASHBOARD:

BAR CHART



STACKED BAR CHART



Set 25: Website Traffic Analysis

R PROGRAM:

```
# Website Traffic Dataset
```

```
date <- c("2023-01-01",
          "2023-01-02",
          "2023-01-03",
          "2023-01-04",
          "2023-01-05")

page_views <- c(1500,1600,1400,1650,1800)

ctr <- c(2.3,2.7,2.0,2.4,2.6)
```

```
# =====
# 2. BAR CHART - TOP N DAYS BY CTR
# =====
```

```
# Sort CTR from highest to lowest
order_ctr <- order(ctr, decreasing=TRUE)
```

```
sorted_date <- date[order_ctr]
sorted_ctr <- ctr[order_ctr]
```

```
# Select Top 3 days
top_n <- 3
```

```
top_date <- sorted_date[1:top_n]
top_ctr <- sorted_ctr[1:top_n]
```

```
barplot(top_ctr,
        names.arg=top_date,
        col="skyblue",
        main="Top 3 Days by Click-through Rate",
        xlab="Date",
        ylab="Click-through Rate (%)")
```

```
# Add values
text(x=seq_along(top_ctr),
     y=top_ctr,
     labels=paste0(top_ctr,"%"),
     pos=3)
```

OUTPUT:

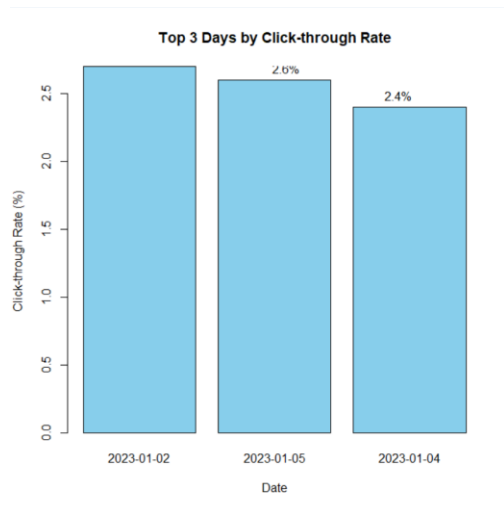
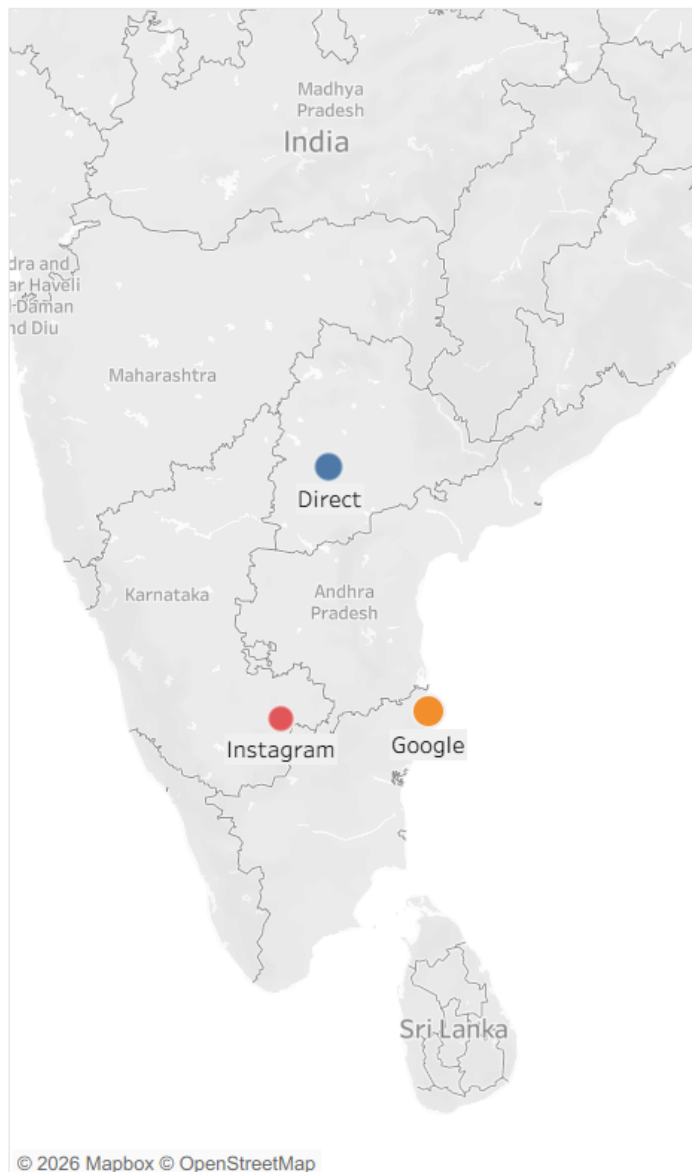


TABLEAU DASHBOARD:

GEO MAP



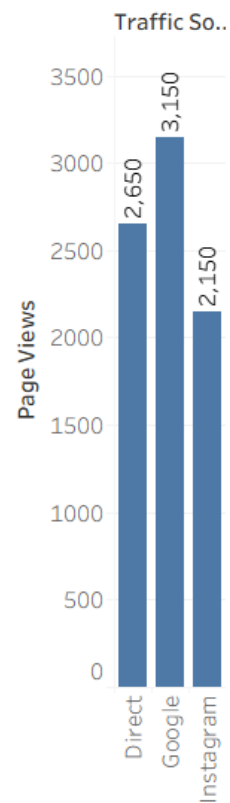
Traffic Source

- Direct
- Google
- Instagram

Page Views

- 2,150
- 2,400
- 2,600
- 2,800
- 3,000
- 3,150

BAR CHART



SET 26: Student Mini Dataset

R PROGRAM:

```
# =====  
# STUDENT MINI DATASET  
# =====  
  
student_id <- c("S01","S02","S03","S04","S05","S06")  
  
gender <- c("Male","Female","Male",  
           "Female","Male","Female")  
  
study_hours <- c(2.0,3.5,1.5,4.0,2.8,3.0)  
  
attendance <- c(78,90,70,95,85,92)  
  
math_score <- c(62,80,55,90,72,82)  
  
science_score <- c(65,85,58,92,74,86)  
  
exam_date <- c("2025-01-10",  
              "2025-01-10",  
              "2025-02-12",  
              "2025-02-12",  
              "2025-03-15",  
              "2025-03-15")  
  
# =====  
# Q1 + Q2 + Q3  
# Display 3 outputs together  
# =====  
  
par(mfrow=c(1,3))  
  
# =====  
# 1. HISTOGRAM + BOXPLOT  
# =====  
  
hist(math_score,  
     col="skyblue",  
     border="white",  
     main="Math Score Distribution",  
     xlab="Math Score",  
     ylab="Frequency")  
  
boxplot(science_score ~ gender,  
        col=c("lightblue","pink"),  
        main="Science Score by Gender",  
        xlab="Gender",  
        ylab="Science Score")  
  
# =====
```

2. SCATTER PLOT

Study Hours vs Math Score

```
# =====
```

```
gender_color <- ifelse(gender=="Male","blue","red")
```

```
plot(study_hours,
     math_score,
     pch=19,
     col=gender_color,
     main="Study Hours vs Math Score",
     xlab="Study Hours",
     ylab="Math Score")
```

Regression line

```
model <- lm(math_score ~ study_hours)
```

```
abline(model,
       col="black",
       lwd=2)
```

```
legend("topleft",
      legend=c("Male","Female","Regression Line"),
      col=c("blue","red","black"),
      pch=c(19,19,NA),
      lty=c(NA,NA,1))
```

```
# =====
```

3. MONTHLY AVERAGE MATH SCORE

```
# =====
```

Convert Exam_Date to Date format

```
exam_date <- as.Date(exam_date,
                    format="%Y-%m-%d")
```

Extract month

```
month <- format(exam_date,
                "%Y-%m")
```

Calculate monthly average

```
monthly_avg <- aggregate(math_score,
                        by=list(Month=month),
                        FUN=mean)
```

```
names(monthly_avg)[2] <- "Average_Math_Score"
```

Display monthly averages

```
print(monthly_avg)
```

Line chart

```
plot(monthly_avg$Average_Math_Score,
     type="o",
     pch=19,
```

```

col="blue",
xaxt="n",
main="Monthly Average Math Score",
xlab="Month",
ylab="Average Math Score")

axis(1,
     at=1:nrow(monthly_avg),
     labels=monthly_avg$Month)

grid()

# Moving average smoothing
moving_avg <- filter(monthly_avg$Average_Math_Score,
                     rep(1/3,3),
                     sides=2)

lines(moving_avg,
      type="o",
      pch=19,
      col="red",
      lwd=2)

legend("topleft",
      legend=c("Monthly Average","Moving Average"),
      col=c("blue","red"),
      lty=1,
      pch=19)

# Reset layout
par(mfrow=c(1,1))

# =====
# INTERPRETATION
# =====

cat("\nQ1 Interpretation:\n")
cat("Math scores show variation among students.\n")
cat("The boxplot compares Science scores between Male and Female students.\n")

cat("\nQ2 Interpretation:\n")
cat("The scatter plot shows the relationship between Study Hours and Math Score.\n")
cat("The regression line indicates the overall direction of the relationship.\n")

cat("\nQ3 Interpretation:\n")
cat("Monthly average Math scores are:\n")
print(monthly_avg)

cat("\nThe monthly average generally increases from January to March.\n")

```

OUTPUT:

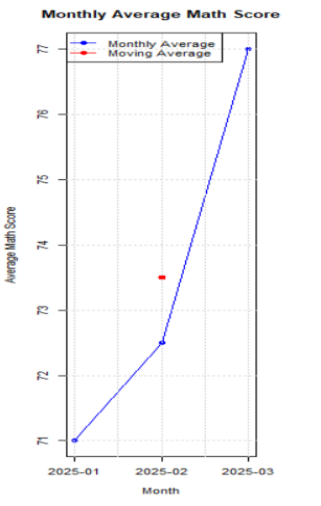
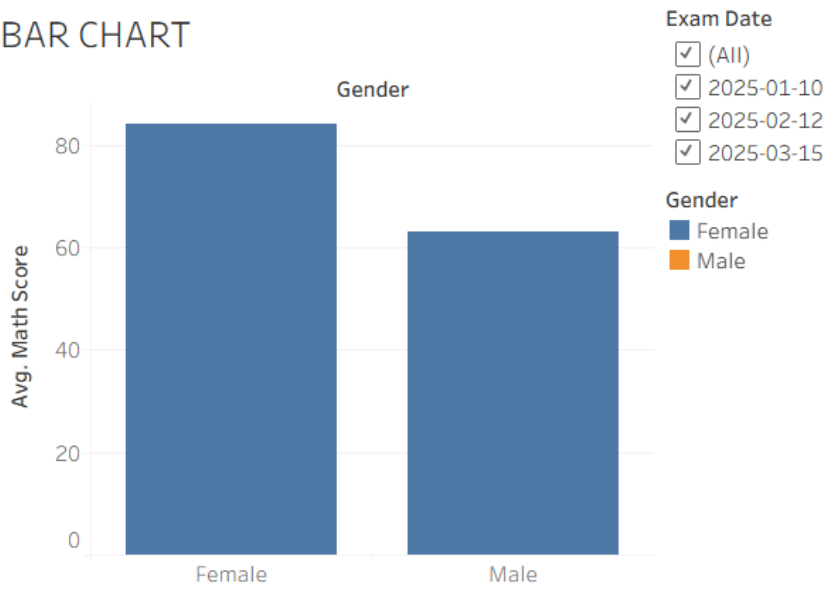
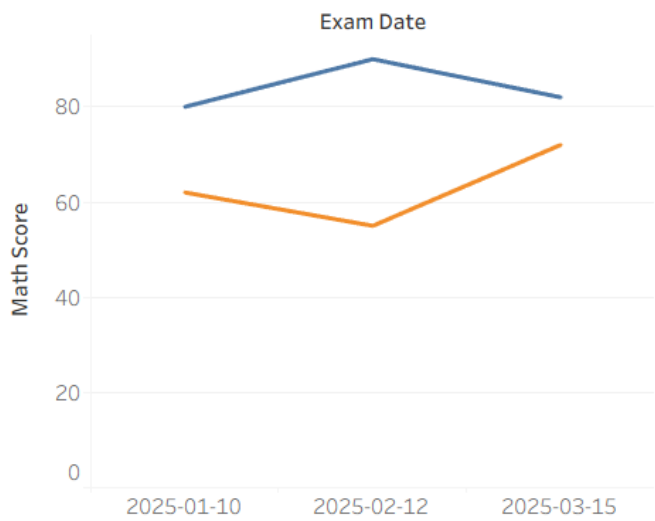


TABLEAU DASHBOARD:

BAR CHART



LINE CHART



SET 27: Patient Health Risk Analysis

R PROGRAM:

```
# =====  
# SET 27 - PATIENT HEALTH RISK ANALYSIS  
# =====  
  
patient_id <- c("P1","P2","P3","P4","P5")  
  
age <- c(25,40,55,35,60)  
  
bmi <- c(22,28,30,26,32)  
  
bp <- c(120,135,145,130,150)  
  
cholesterol <- c(180,210,240,200,260)  
  
# =====  
# Q1 - SCATTERPLOT MATRIX  
# =====  
  
health_data <- data.frame(  
  Age=age,  
  BMI=bmi,  
  BP=bp,  
  Cholesterol=cholesterol  
)  
  
pairs(health_data,  
  main="Scatterplot Matrix of Health Indicators",  
  pch=19,  
  col="blue")
```

OUTPUT:

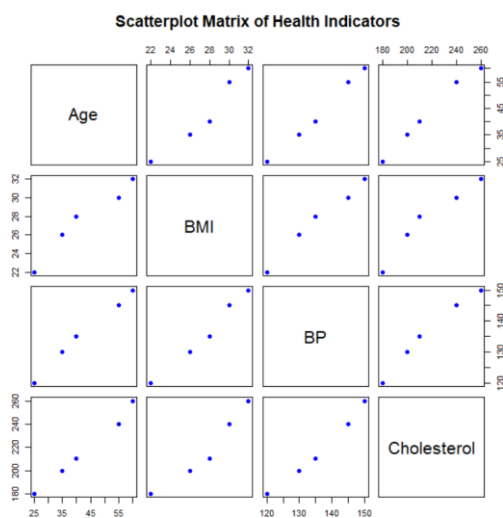
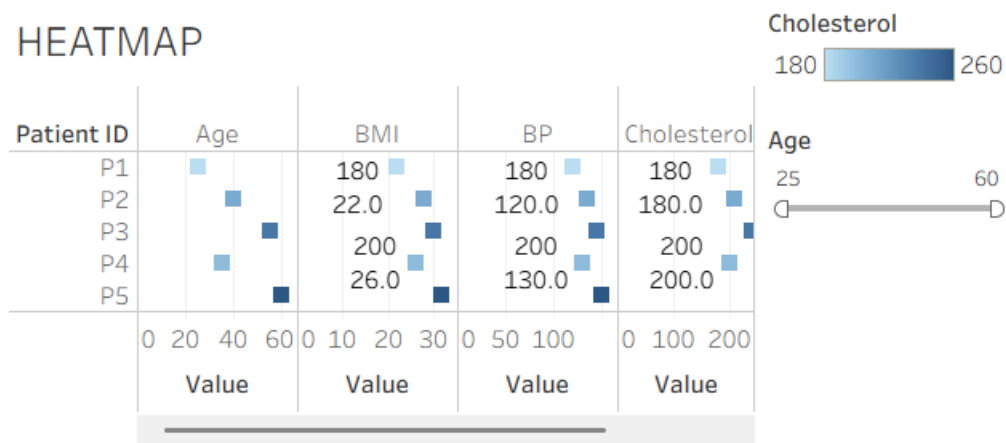


TABLEAU DASHBOARD:

HEATMAP



SET 28. Vehicle Performance Analysis

R PROGRAM:

```
# =====
# SET 28 - VEHICLE PERFORMANCE ANALYSIS
# =====

vehicle_id <- c("V1","V2","V3","V4","V5")

engine_size <- c(1.5,2.0,3.0,2.5,1.8)

horsepower <- c(110,150,250,200,130)

fuel_efficiency <- c(18,15,12,14,17)

top_speed <- c(180,200,250,220,190)

safety_rating <- c(4,5,5,4,3)

# =====
# Q1 - VIOLIN PLOT
# =====

# Install once if required:
# install.packages("vioplot")

library(vioplot)

vioplot(
  fuel_efficiency[safety_rating == 3],
  fuel_efficiency[safety_rating == 4],
  fuel_efficiency[safety_rating == 5],
```

```

names=c("Safety 3","Safety 4","Safety 5"),
col=c("skyblue","lightgreen","orange"),
main="Fuel Efficiency by Safety Rating",
xlab="Safety Rating",
ylab="Fuel Efficiency (km/l)"
)

# Add individual observations
points(
  c(1,2,3)[match(safety_rating,
                  c(3,4,5))],
  fuel_efficiency,
  pch=19
)

```

OUTPUT:

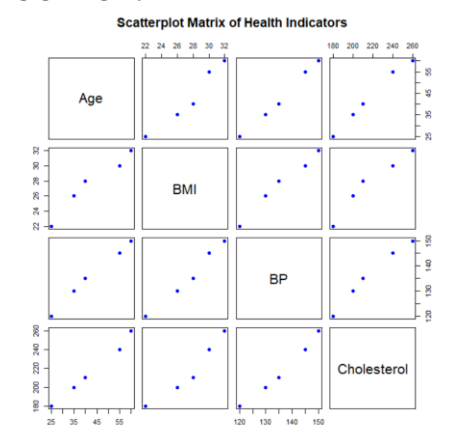
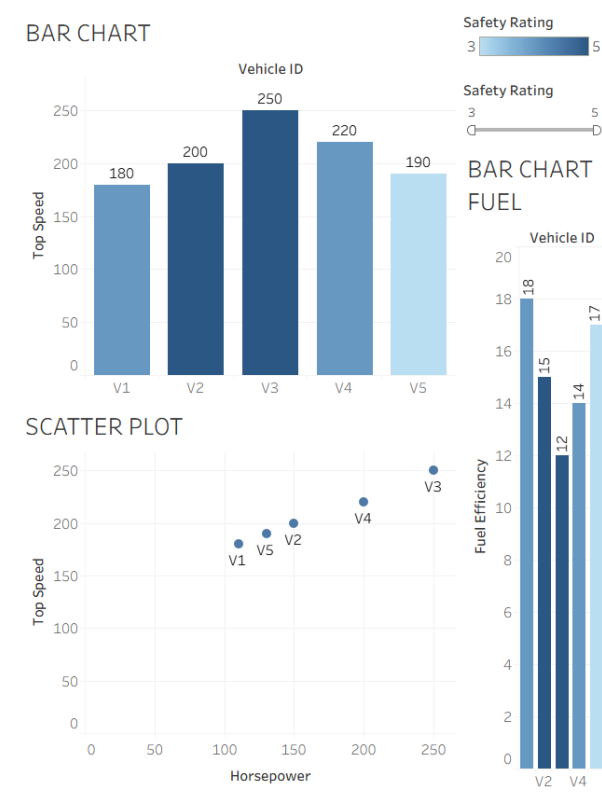


TABLEAU DASHBOARD:



SET 29 .Student Academic Performance

R PROGRAM:

```
# =====  
# SET 29 - STUDENT ACADEMIC PERFORMANCE  
# Q1 + Q2 + Q3 IN ONE R PROGRAM  
# =====  
  
# Dataset  
student_id <- c("S1","S2","S3","S4","S5")  
  
age <- c(19,21,20,22,23)  
  
study_hours <- c(12,8,15,10,7)  
  
attendance <- c(90,70,95,85,60)  
  
test_score <- c(85,70,92,80,65)  
  
participation_score <- c(8,7,9,8,6)  
  
# =====  
# DISPLAY ALL 3 QUESTIONS TOGETHER  
# =====  
  
par(mfrow=c(1,3))  
  
# =====  
# Q1 - STACKED AREA CHART  
# Test Score + Participation Score  
# =====  
  
x <- 1:length(student_id)  
  
total_score <- test_score + participation_score  
  
plot(x,  
     total_score,  
     type="n",  
     xaxt="n",  
     ylim=c(0,max(total_score)),  
     main="Test & Participation Scores",  
     xlab="Student",  
     ylab="Score")  
  
axis(1,  
     at=x,  
     labels=student_id)  
  
# Test Score area  
polygon(  
  c(x, rev(x)),  
  c(rep(0,length(x)), rev(test_score)),
```

```

col="skyblue",
border="blue"
)

# Participation Score area
polygon(
  c(x, rev(x)),
  c(test_score, rev(total_score)),
  col="lightgreen",
  border="darkgreen"
)

legend(
  "topleft",
  legend=c("Test Score",
            "Participation Score"),
  fill=c("skyblue","lightgreen")
)

grid()

```

```

# =====
# Q2 - BOXPLOT
# Study Hours grouped by Attendance Quartiles
# =====

```

```

# Calculate attendance quartiles
attendance_quartiles <- quantile(
  attendance,
  probs=c(0,0.25,0.50,0.75,1)
)

```

```

print(attendance_quartiles)

```

```

# Create attendance groups
attendance_group <- cut(
  attendance,
  breaks=attendance_quartiles,
  include.lowest=TRUE,
  labels=c("Q1","Q2","Q3","Q4")
)

```

```

print(attendance_group)

```

```

boxplot(
  study_hours ~ attendance_group,
  col=c("skyblue",
        "lightgreen",
        "orange",
        "pink"),
  main="Study Hours by Attendance Quartile",
  xlab="Attendance Quartile",
  ylab="Study Hours"
)

```

```
grid()
```

```
# =====  
# Q3 - DENSITY PLOT  
# Test Score Distribution  
# =====
```

```
test_density <- density(test_score)
```

```
plot(  
  test_density,  
  col="blue",  
  lwd=2,  
  main="Density Plot of Test Scores",  
  xlab="Test Score",  
  ylab="Density"  
)
```

```
# Shade density area
```

```
polygon(  
  test_density,  
  col=rgb(0,0,1,0.2),  
  border="blue"  
)
```

```
# Show actual observations
```

```
rug(  
  test_score,  
  col="red"  
)
```

```
grid()
```

```
# =====  
# RESET GRAPH LAYOUT  
# =====
```

```
par(mfrow=c(1,1))
```

```
# =====  
# DISPLAY RESULTS  
# =====
```

```
cat("\n=====\\n")  
cat("STUDENT ACADEMIC PERFORMANCE ANALYSIS\\n")  
cat("=====\\n")
```

```
cat("\nTest Scores:\\n")  
print(test_score)
```

```
cat("\nParticipation Scores:\\n")
```

```

print(participation_score)

cat("\nAttendance Quartiles:\n")
print(attendance_quartiles)

cat("\nAttendance Groups:\n")
print(attendance_group)

cat("\nStudy Hours:\n")
print(study_hours)

cat("\n=====")
cat("INTERPRETATION\n")
cat("=====")

cat("\nQ1: Test scores and participation scores vary across students.\n")
cat("S3 has the highest test score and participation score.\n")

cat("\nQ2: Study hours are compared across attendance quartiles.\n")
cat("The small sample size means the quartile-level boxplots should\n")
cat("be interpreted cautiously.\n")

cat("\nQ3: The density plot shows the distribution of test scores.\n")
cat("The scores range from 65 to 92 with no obvious extreme outlier.\n")

```

OUTPUT:

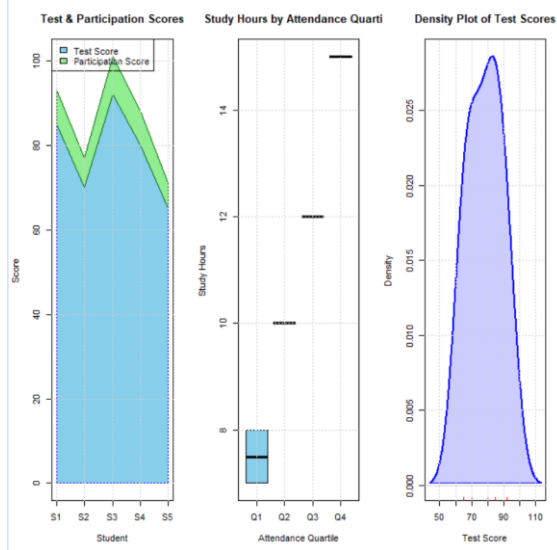
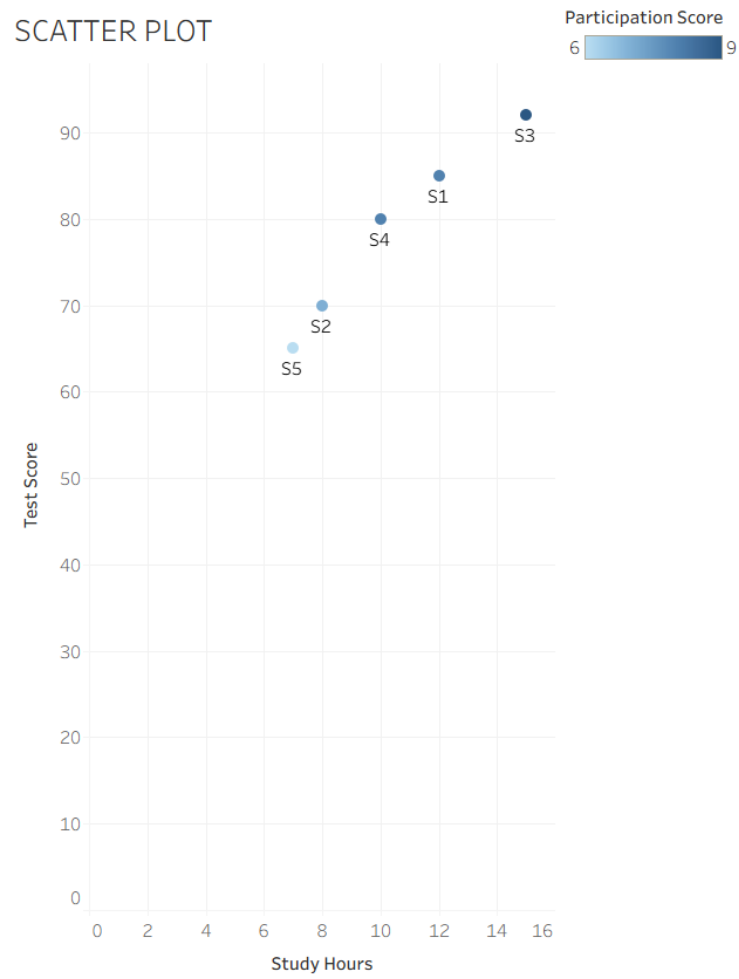


TABLEAU DASHBOARD:

SCATTER PLOT



SET 30: Mobile App usage Dataset

R PORGRAM:

```
# =====  
# SET 30 - MOBILE APP USAGE ANALYSIS  
# Q1 + Q2 + Q3 IN ONE R PROGRAM  
# =====  
  
# =====  
# DATASET  
# =====  
  
user_id <- c("U01","U02","U03","U04","U05","U06")  
  
gender <- c("Male","Female","Male",  
           "Female","Male","Female")  
  
age <- c(20,22,19,21,23,20)  
  
screen_time <- c(4.5,6.0,3.2,7.1,2.8,5.4)  
  
app_usage <- c(18,25,12,30,10,22)  
  
data_used <- c(2.4,3.8,1.6,4.5,1.2,3.1)  
  
satisfaction <- c(3,5,3,5,2,4)  
  
usage_date <- c("2025-01-08",  
               "2025-01-08",  
               "2025-02-11",  
               "2025-02-11",  
               "2025-03-14",  
               "2025-03-14")  
  
# =====  
# Q1 - HISTOGRAM AND DENSITY PLOT  
# Screen Time Distribution  
# =====  
  
hist(screen_time,  
     probability=TRUE,  
     col="skyblue",  
     border="white",  
     main="Screen Time Distribution",  
     xlab="Screen Time (hrs)",  
     ylab="Density")  
  
# Add density curve  
lines(density(screen_time),  
      col="red",  
      lwd=2)
```

```

legend("topright",
      legend=c("Histogram",
               "Density Curve"),
      fill=c("skyblue",NA),
      border=c("white",NA),
      col=c(NA,"red"),
      lty=c(NA,1),
      lwd=c(NA,2))

```

```

# =====
# Q2 - SCATTER PLOT
# Data Used vs Screen Time
# =====

```

```

plot(data_used,
      screen_time,
      pch=19,
      col="blue",
      main="Data Used vs Screen Time",
      xlab="Data Used (GB)",
      ylab="Screen Time (hrs)")

```

```

# Add trend line
model <- lm(screen_time ~ data_used)

```

```

abline(model,
      col="red",
      lwd=2)

```

```

# Calculate correlation
correlation <- cor(data_used,
                  screen_time)

```

```

cat("\nCorrelation between Data Used and Screen Time:",
    round(correlation,3))

```

```

legend("topleft",
      legend=c("Users",
               "Trend Line"),
      col=c("blue","red"),
      pch=c(19,NA),
      lty=c(NA,1))

```

```

# =====
# Q3 - AVERAGE SATISFACTION BY GENDER
# =====

```

```

average_satisfaction <- tapply(
  satisfaction,
  gender,
  mean
)

```

```

print(average_satisfaction)

# Bar chart
bar_positions <- barplot(
  average_satisfaction,
  col=c("lightblue","pink"),
  main="Average Satisfaction Score by Gender",
  xlab="Gender",
  ylab="Average Satisfaction Score",
  ylim=c(0,5)
)

# Add value labels
text(
  x=bar_positions,
  y=average_satisfaction,
  labels=round(average_satisfaction,2),
  pos=3
)

# =====
# DISPLAY RESULTS
# =====

cat("\n===== \n")
cat("MOBILE APP USAGE ANALYSIS\n")
cat("===== \n")

cat("\nQ1 - Screen Time Values:\n")
print(screen_time)

cat("\nQ2 - Correlation:\n")
print(round(correlation,3))

cat("\nQ3 - Average Satisfaction by Gender:\n")
print(round(average_satisfaction,2))

# =====
# INTERPRETATION
# =====

cat("\n===== \n")
cat("INTERPRETATION\n")
cat("===== \n")

cat("\nQ1:\n")
cat("Screen time ranges from 2.8 to 7.1 hours.\n")
cat("The histogram and density curve show the distribution\n")
cat("of screen-time values among the users.\n")

cat("\nQ2:\n")
cat("The correlation measures the relationship between\n")

```

```
cat("data usage and screen time.\n")

cat("\nQ3:\n")
cat("Female users have a higher average satisfaction score\n")
cat("than male users in this dataset.\n")
```

OUTPUT:

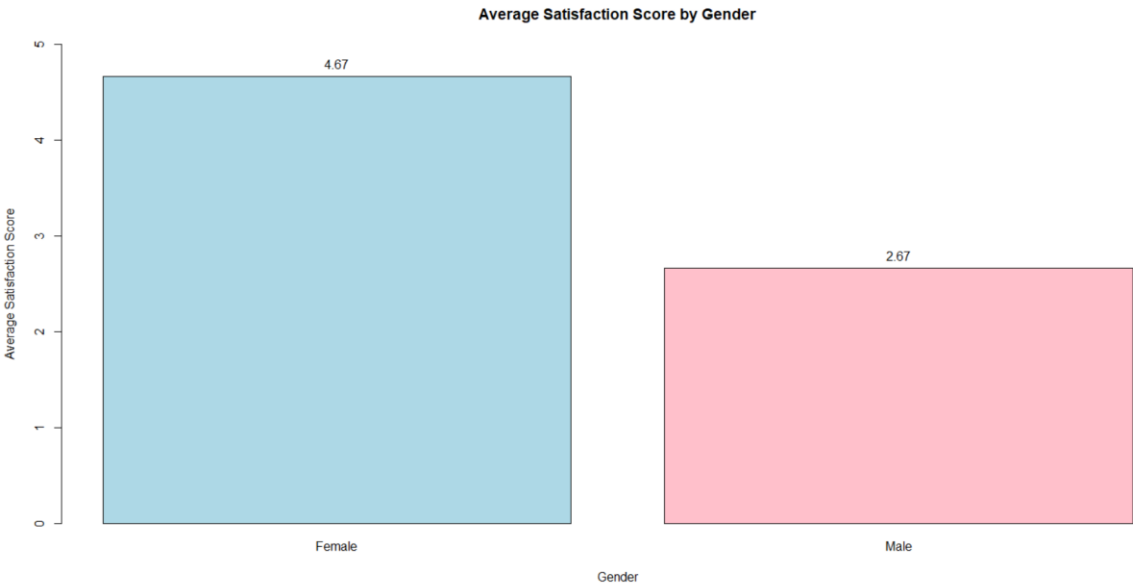


TABLEAU DASHBOARD:

